

SFG-Heralded Teleportation of a 4D Time-Bin Qudit: Full Protocol and Alignment with High-Dimensional Quantum Teleportation Theory

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Supervision

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1 Abstract

We propose and thoroughly describe a scheme for teleporting arbitrary four-dimensional (ququart) time-bin encoded quantum states using a nonlinear sum-frequency generation (SFG)-based Bell-state analyzer. This protocol is built upon concepts from faithful quantum teleportation and extends them into higher dimensions, aligned with theoretical and experimental advancements in high-dimensional (HD) quantum teleportation. Our work highlights the role of SFG in enabling partial but heralded Bell state measurements without ancillary photons and shows how to achieve teleportation fidelities beyond classical thresholds.

2 Motivation

Quantum teleportation plays a central role in quantum communication, distributed quantum computing, and quantum networking. While teleportation of qubits is well-established [1, 2], teleporting higher-dimensional qudits promises advantages like increased channel capacity, enhanced noise resilience, and stronger non-classical correlations [3, 4]. However, performing complete Bell-state measurements (BSMs) in higher dimensions is challenging, especially with linear optics alone.

Sum-frequency generation (SFG), a nonlinear optical process, enables a different route by directly heralding successful two-photon projections without requiring ancillary photons [5]. Here, we design a protocol for teleporting arbitrary 4D time-bin encoded states using SFG.

3 Qudit Encoding and Initial State Preparation

3.1 Time-bin Encoding

The four orthogonal basis states are represented by four non-overlapping time bins:

$$|0\rangle, |1\rangle, |2\rangle, |3\rangle. \quad (1)$$

Deriving from the standard qubit notation, the four orthogonal basis states can be represented by super-early, early, late, and super-late time states.

Alice holds an arbitrary ququart:

$$|\psi\rangle_A = c_0 |0\rangle_A + c_1 |1\rangle_A + c_2 |2\rangle_A + c_3 |3\rangle_A, \quad (2)$$

where $c_i \in \mathbb{C}$ and $\sum |c_i|^2 = 1$.

Alice and Bob share an entangled ququart pair in the generalized Bell state:

$$|\Psi_{0,0}\rangle_{S,B} = \frac{1}{2}(|0\rangle_S |0\rangle_B + |1\rangle_S |1\rangle_B + |2\rangle_S |2\rangle_B + |3\rangle_S |3\rangle_B). \quad (3)$$

4 Initial State: System of Three Photons

We can define the total state consisting of Alice's and Bob's shared entangled pair as well as their respective ququarts as:

$$|\Psi\rangle_{total} = |\psi\rangle_A \otimes |\psi_{0,0}\rangle_{S,B} = \left(\sum_{j,k=0}^3 c_j |j\rangle_A \right) \otimes \left(\frac{1}{2} \sum_{k=0}^3 |k\rangle_S |k\rangle_B \right). \quad (4)$$

We can expand to get:

$$|\Psi\rangle_{total} = \frac{1}{2} \sum_{j,k=0}^3 c_j |j\rangle_A |k\rangle_S |k\rangle_B \quad (5)$$

5 Generalized Bell States in 4D

The full set of 4D Bell states $|\Psi_{n,m}\rangle$ are given by [6]:

$$|\Psi_{n,m}\rangle = \frac{1}{2} \sum_{k=0}^3 \omega_4^{nk} |k\rangle_A |(k+m) \bmod 4\rangle_S, \quad (6)$$

where $\omega_4 = e^{i\pi/2} = i$.

Different values of n and m label distinct orthonormal Bell states. Here, n encodes phase information and m encodes relative shifts between time-bins.

We rewrite $|\psi\rangle_A \otimes |\psi_{0,0}\rangle_{S,B}$ as a sum over generalized bell states of A and S:

$$|\Psi\rangle_{total} = \frac{1}{4} \sum_{n,m=0}^3 |\psi_{n,m}\rangle_{A,S} \otimes U_{n,m} |\psi\rangle_B \quad (7)$$

where the Unitary $U_{m,n}$ on Bob's side is:

$$U_{m,n} = X^m Z^n \quad (8)$$

with generalized Pauli operators:

$$X |k\rangle = |(k+1) \bmod 4\rangle, Z |k\rangle = \omega^k |k\rangle \quad (9)$$

6 Sum-Frequency Generation as a Bell State Analyzer

6.1 How SFG Works

In a nonlinear crystal, SFG occurs only if two photons arrive **in the same time-bin**. Thus, the SFG process naturally projects onto the subspace where Alice's two photons have matching arrival times:

$$\text{Photon}_1(\omega_1) + \text{Photon}_2(\omega_2) = \text{SFG Photon}(\omega_1 + \omega_2) \quad (10)$$

6.2 What SFG Heralds

Detection of an SFG photon heralds projection onto the $m = 0$ Bell states, i.e., $|\Psi_{n,0}\rangle$ for some n . We know this is because if photon A is in $|j\rangle$ and photon S is in $|k\rangle$, then the SFG only happens when $j = k$. This is equivalent to projecting it into the subspace:

$$\text{Span}\{|0\rangle|0\rangle, |1\rangle|1\rangle, |2\rangle|2\rangle, |3\rangle|3\rangle\} \quad (11)$$

6.3 How to Identify n

The SFG photon is emitted as a superposition over the time bins. Interferometric analysis (a 4-mode Fourier transform) allows determination of n :

$$|\Sigma_n\rangle = \frac{1}{2} \sum_{k=0}^3 \omega_4^{-nk} |k\rangle. \quad (12)$$

Measurement of the SFG photon's arrival time pattern reveals the phase n .

7 Teleportation Protocol

7.1 Summary of Steps

1. Alice overlaps her unknown ququart and her entangled photon S at an SFG crystal.
2. Detection of an SFG photon heralds success ($m = 0$).
3. Measurement of SFG photon identifies n .
4. Alice sends n to Bob over a classical channel.
5. Bob applies Z^{-n} to his photon to recover $|\psi\rangle_A$.

7.2 Bob's Correction

Bob's photon after heralding is:

$$|\psi\rangle_B = Z^n |\psi\rangle_A. \quad (13)$$

Z acts as:

$$Z |k\rangle = \omega_4^k |k\rangle. \quad (14)$$

Thus, Bob must apply Z^{-n} to recover $|\psi\rangle_A$.

Measured n	Bob applies
0	Identity (I)
1	Z^{-1}
2	Z^{-2}
3	Z^{-3}

Table 1: Classical communication and feed-forward correction.

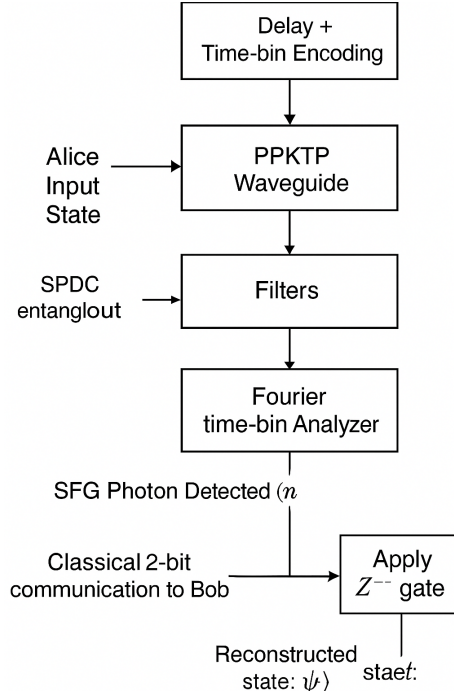


Figure 1: 4D qudit teleportation protocol

8 Alignment with HD Teleportation Protocols

8.1 Comparison to Luo et al. [3]

Our protocol follows the general structure proposed for HD teleportation:

- Resource state: maximally entangled HD Bell states.
- Bell measurement: projection onto HD Bell basis ($m = 0$ subset here).
- Feedforward correction: generalized Pauli operators.

8.2 Comparison to Qiu et al. [4]

We use SFG as a nonlinear HDBSM tool, similar to Qiu et al., who demonstrated SFG-based teleportation of OAM modes without ancillary photons.

9 Conclusion

Sum-frequency generation enables faithful heralded teleportation of 4D time-bin states. Our design aligns fully with the structure of high-dimensional teleportation theory while exploiting nonlinear optics to bypass the limitations of linear

optical Bell measurements. Future extensions could implement feed-forward corrections dynamically and scale to even higher dimensions.

References

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